

- Removal of host plants
- Installation of yellow sticky traps
- Spray application of imidacloprid (0.0036%) or triazophos (0.04%) during heavy infestation.
- Spray neem oil 3% or NSKE 5%
- Release of predators viz., Coccinellid predator, *Cryptolaemus montrouzieri*
- Release of parasitoids viz., *Encarsia haitierrsis* and *E.guadeloupae*

INSECT-PESTS OF TEMPERATE FRUITS

Sanjose scale	Apple shot hole borer	Hairy caterpillars	Apple leaf blotch miner
Woolly apple aphid	Apple stem borer	Thrips	Apple leaf hoppers
Apple root borer	Apple fruit moth	Apple fruit rhynhites	Apricot fruit moth

1. Sanjose scale *Quadraspidiotus perniciosus*

- **Order:** Homoptera
- **Family:** Diaspidae
- **Host:** 200 species, Rosaceae family (apple, pear, peach, plum etc.)
- ❖ **Symptoms:**
 - Less infested trees show small, grey specks on the bark surface.
 - Severely infested trees have the bark covered with grey layer of over-lapping scales appearing as if sprayed wood ash.
 - Red inflamed areas on fruits.
- ❖ **Nature of Damage**
 - **Suck** cell sap, as a result , the young plants in the nursery become weak and ultimately die.
 - Stems and branches of trees have poor growth, low vigor and ultimately death owing to sucking sap

- Infested fruits have more scales at the stem end and the calyx
- Marketing value of the fruits is greatly hampered due to infestation

❖ **Management**

- Spray between 1/2" green stage and tight cluster stage with TSO
- Last week of May-First week of June spray 0.04% chlorpyrifos/0.025% Metasystox/Summer oil
- Post-harvest spray oil 0.04% chlorpyrifos

2. Woolly apple aphid: *Eriosoma lanigerum*

- **Order: Homoptera**
- **Family: Aphididae**
- **Host: Apple, crab apple**
- Aerial parts-white woolly mass
- Roots-gall formation

❖ **Nature of Damage**

- Sucking sp from stems, twigs and roots.
- Affected plants-stunted growth, loss of vigor, reducing fruiting and gall formation
- Severe galls in the roots cause small size fruits and early coloration
- Pest over winters as nymphs in crevices, loose bark and underground parts
- Reproduction parthenogenetically

❖ **Management**

- Non-Bearing Trees: Phorate 10-30g or carbofuran 30-50g at 5cm depth of soil
- Bearing Trees: Spray in June-October 0.04% chlorpyrifos or 0.08% triazophos
- Drench same during April and Sep/Oct
- Apply 25-30g phorate or 70-80g carbofuran
- *Aphelinus mali* Biocontrol agent
- Remove water shoots
- M-25 rootstock resistant
- Malling Merton series resistant (Kotkhai research)

COCONUT

1. Rhinoceros beetle: *Oryctes rhinoceros*

❖ **Symptoms of damage**

- Central spindle appears cut or toppled

- Fully opened fronds showing characteristic diamond shaped cuttings
- Holes with chewed fibre sticking out at the base of central spindle.

❖ **Management**

- Remove and burn all dead coconut trees in the garden (which are likely to serve as breeding ground) to maintain good sanitation.
- Incorporate the entomopathogen i.e., fungus (*Metarrhizium anisopliae*) in manure pits to check the perpetuation of the pest.
- Soak castor cake at 1 kg in 5 l of water in small mud pots and keep them in the coconut gardens to attract and kill the adults.
- For seedlings, apply 3 naphthalene balls/palm weighing 3.5 g each at the base of inter space in leaf sheath in the 3 inner most leaves of the crown once in 45 days.
- Set up light traps following the first rains in summer and monsoon period to attract and kill the adult beetles.
- Field release of Baculovirus inoculated adult rhinoceros beetle @ 15/ha reduces the leaf and crown damage caused by this beetle.
- Apply mixture of either neem seed powder + sand (1:2) @150 g per palm or neem seed kernel powder + sand (1:2) @150 g per palm in the base of the 3 inner most leaves in the crown.
- Place phorate 10 G 5 g in perforated sachets in two inner most leaf axils for 2 times at 6 months intervals.
- Set up rhinolure pheromone trap @ 1 per ha to trap and kill the beetles.
- Maintain good sanitation soak castor cake at 1 kg in 5 litres of water in small mud pots and keep them in coconut gardens to attack and kill adults.
- Apply 3 Naphthalene balls /palm weighing 3.5g each at the base of the inter space.

2. Red palm weevil: *Rhynchophorus ferrugineus*

❖ **Symptoms of damage**

- Holes on trunk with brownish ooze.

- Yellowing of inner leaves.
- Gradual wilting of central shoot in the crown.
- Remove and burn all wilting or damaged palms in coconut gardens to prevent further perpetuation of the pest.
- Avoid injuries on stems of palms as the wounds may serve as oviposition sites for the weevil. Fill all holes in the stem with cement.
- Avoid the cutting of green leaves. If needed, they should be cut about 120 cm away from the stem.
- Fill the crown and the axils of topmost three leaves with a mixture of fine sand and neem seed powder or neem seed kernel powder (2:1) or lindane 1.3 D (1:1 by volume) once in three months to prevent the attack of rhinoceros beetle damage in which the red palm weevil lays eggs.
- Plug all holes and inject pyrocone E or carbaryl 1% or 10 ml of monocrotophos into the stem by drilling a hole above the points of attack.
- Setting up of attractant traps (mud pots) containing sugarcane molasses 2½ kg or toddy 2½ litres + acetic acid 5 ml + yeast 5 g + longitudinally split tender coconut stem/logs of green petiole of leaves of 30 numbers in one acre to trap adult red palm weevils in large numbers.
- Install pheromone trap @ 12 per ha

3. Black headed caterpillar: *Opisina arenosella*

❖ Symptoms of Damage

- Dried up patches on leaflets of the lower leaves
- Galleries of silk and frass on underside of leaflets

❖ Management

- Among the larval parasitoids, the bethylid *Goniozus nephantidis* is the most effective in controlling the pest.
- The optimum level of release is 1:8 of host-parasitoid ratio.

- The parasitoid should be released @3000/ha under the coconut trees when the pest is in the 2nd or 3rd instar larval stage. Parasitoid release trap may be used to release the parasitoid at the site of feeding.
- Parasitoids should not be released in the crown region since they will be killed by predators like spiders and reduviid bugs.
- Remove and burn all affected leaves/leaflets.
- Spray malathion 50 EC 0.05% (1ml/lit) to cover the undersurface of the leaves thoroughly in case of severe epidemic outbreak of the pest in young palms.

TEA

1. Tea mosquitobug, *Helopeltis theivora*

- It is considered to be a serious pest of a tea.
- Adults and nymphs suck the sap from buds, young leaves and tender stems.
- Puncturing the plant tissues with their needle like rostrum and inject toxic saliva.
- Punctures appear as reddish-brown spots.
- Leaves curl up, badly deformed and shoots dry up.

2. Tea Mites

Common Name	Scientific Name
Red spider mite	Oligonychus coffeae
Scarlet mite	Brevipalpus californicus
Purple mite	Calacarus carinatus
Pink mite (or) Orange mite	Acaphylla theae
Yellow mite	Polyphagotarsonemus latus

❖ Management

- Spray dicofol 18EC 2 ml/lit or ethion 50 EC 2 ml/lit or monocrotophos 1ml/lit
- Application of wettable sulphur 80 WP 2g/lit using hand operated sprayer.

1. Coffee berry borer, *Hypothenemus hampei*

- In India, the pest was first noticed on a few plantations in the Nilgiris during early 1990.
- Pin hole at the tip of the berries (novel region)
- Severe infestation - two or more holes may be seen.
- Female beetle bores into the berries through the navel region make tunneling and feed inside content.
- Powdery substance pushed out through the holes

❖ Management

- Proper adoption of cultural practices and phytosanitary measures important for management of coffee berry borer.
- Transportation of infested coffee to uninfected areas is the main reason for spread.
- Crop bags should be fumigated before delivery to estates to avoid cross infestation.
- Timely harvest
- Gleaning - spreading gunny bags or polythene sheets on the ground after picking the berries - minimize gleaning.
- Maintain optimum shade and good drainage.
- Dipping infested berries in boiling water for 2-3 minutes kills all the stages inside.
- Drying of coffee beans – prevents breeding of beetles in stored coffee based on moisture content level.
- Arabica (10% moisture content) - 16kg/lit
- Roubsta (11% moisture content) - 18kg/lit
- Install attractant trap to collect and kill the beetles

SOME IMPORTANT INSECTS OF MAJOR CROPS

Rice		
Stem borers	Yellow stem borer	Scirpophaga incertulas
	Pale headed borer	Chilo suppressalis
	Dark headed striped borer	Chilo polychrysa
	Pink stem borer	Sesamia inferens
	Sugarcane shoot borer	Chilo infuscatellus
Leaf feeders	Rice gall midge	Orseolia oryzae
	Rice caseworm	Nymphula depunctalis
	Rice ear cutting caterpillar	Mythimna separata
	Rice hispa	Diclodispa armigera
	rice grasshopper	Hieroglyphus banian
Sap suckers	rice bug	Leptocoriosa acuta